

University of Chester

M.Sc. Sport Performance Analysis

SS7316 Research Report

Title:

From Strokes to Seconds: Principal Components and
Predictive Factors of the Men's 4 x 100m Freestyle Relay

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Introduction

Understanding the relationship between performance metrics and their impacts on a sporting event is a fundamental aim of performance analysts (M. D. Hughes & Bartlett, 2002). Highlighting important metrics within sport can provide coaches and athletes with important benchmarks that can be utilized for peer-to-peer comparisons and training strategies. Performance metrics in swimming are unique in their practical utility but are often interrelated based on their definition (Staunton et al., 2024a). For example, free swimming speed (FSS) consists of distance per stroke (DPS) and stroke rate (SR) during the middle section of the pool, so any change in DPS or SR would impact FSS. The dimensionality of swim race data creates a multivariate dataset that makes it difficult to differentiate relative value in terms of overall performance. This characteristic makes dimensionality reduction techniques such as principal component analysis (PCA) appropriate as they are able to reduce dataset size and complexity while preserving as much relevant information as possible (Sewell, 2007).

PCA acts by transforming the dataset into a new coordinate system whose axes (the principal components) are linear combinations of the original variables. Each axis is chosen to be uncorrelated (orthogonal) to the prior and capture the maximum remaining variance. Axes are chosen by performing an eigenvalue-decomposition of the data's covariance matrix. Each component's eigenvector (or loading) gives the weights of the original variables. The corresponding eigenvalue indicates how much variance that component explains. The principal component that captures the most variation within the dataset is known as PC1. The following principal component (PC2), being uncorrelated with PC1, has the second highest loading. Components are retained using the Kaiser criterion (eigenvalue > 1) and inspection of the scree plot. PCA can reveal groupings of variables that tend to vary together, showing themes or factors of a dataset that can be more easily understood by coaches and athletes

(Weaving et al., 2019). For each retained component, the variable with the highest loading will then be retained to be used in a stepwise linear regression. Stepwise linear regression operates systematically by entering a variable into the regression formula using an entry/removal criterion.

Previous research using similar procedures has been done on international level short-course races in both men and women, so any additional research on long-course or relay events would be novel (Staunton et al., 2024a, 2024b). Previous prediction models formed for the 4 x 200m freestyle event have used qualitative variables such as world ranking, pacing strategy, start lap strategy, and relay leg order as indicators of overall performance (Wu et al., 2021). While this information is valuable in certain circumstances, predicting a relay team's placing based off their current world standings provides limited insight into how they achieved their world standings in the first place or how they could improve their performance.

The men's 4 x 100m freestyle relay is the fastest of all relays and is well known for its close finishes. The 2024 Olympic relay final had an average margin between placings of 0.55 seconds, highlighting how every millisecond matters — especially when any small difference can be compounded across a team of 4. Given the importance of this event, and the limited amount of research in the area; this proposed study aims to record relevant continuous variables (ex. distance per stroke, breakout distance, turn time, etc.) within the men's 4 x 100m freestyle event, conduct a statistical analysis using PCA and a stepwise linear regression to produce a prediction model based on the resulting regression equation. The model will then be tested via cross-fold validation and evaluated against out of sample data. The aim is to create a prediction model that uses the most impactful variables to predict total race time. This model could then serve as a training tool for coaches and athletes, similar to that of previous research (Staunton et al., 2024b, 2024a; Wu et al., 2021).

Methods

Participants

This study included 80 male athletes who competed in the 4 x 100m freestyle relay finals at the 2023 World Aquatic Championships, the 2024 Olympic Games, or the 2025 World Aquatic Championships. The starting athlete in a relay team is the only team member with a standard start position. Each subsequent athlete after the starter, implements a flying start, also known as a relay takeover. Previous research has shown differences in emersion time, time to 15m, and reaction time between an athletes' relay takeover start and their individual start (Qiu & Calvo, 2020). As such, the sample was separated into two groups with 20 for the “starters” and 80 for the “non-starters”.

Data Collection

Each athlete in the selected 4 x 100m freestyle events were analysed in a bespoke program (NEMO) using previously recorded video footage. It should be noted that only analysts certified within the AquaticsGB program are able to use this program. Additionally, NEMO has an intra-analyst error flagging system that compares newly entered data to data of its peer group based on competition level. Any metric that is outside the groups' mean by more than 1 standard deviation is flagged for an analyst to review given the context of the athlete's position within the competition level. 12 variables were gleaned from the video analysis, details shown below on *Table 1. Analysis Definitions*. All location-time variables were recorded when the centre of an athletes' head passed the appropriate distance. Using a calibrated map of the pool(s), distances were calculated via the coloured rungs of the lane ropes to approximately 0.1m; detailed examples can be found in appendix *Section 1. Video Analysis Definitions*. Intra-analyst reliability was assessed using 2 randomly selected teams (8

athletes) from each competition. Analyses were conducted on separate days to minimize any familiarity effects. Percent error was calculated to determine the differences between repeated analysis', resulting in an overall average of $0.55 \pm 1.14\%$ (M. Hughes, 2004). A complete breakdown is provided in appendix *Section 2. Reliability Testing*.

Start Breakout Time	The time at which the athlete's arm leaves the water for the first stroke
Start15	The time at which the athlete reaches the 15m mark
Start15to25	The time difference from the 25m to the 15m mark
In5	Time from the 45m mark to the wall at the end of the first length
Rotate	Time from the last hand entry to when the athlete touches the wall
Turn Breakout Time	The time at which the athlete surfaces after the turn
Out15	Time from the turn to the 15m mark
Out15to25	The time difference from the returning 25m to the 15m mark
Stroke Count	Number of strokes in the length
Stroke Rate*	Number of strokes per minute
Distance Per Stroke*	Amount of strokes / distance covered
Free Swimming Speed*	Distance covered / amount of time in the given distance

Table 1. Analysis Definitions

** Measured as an average from the 15-45m marks of both lengths*

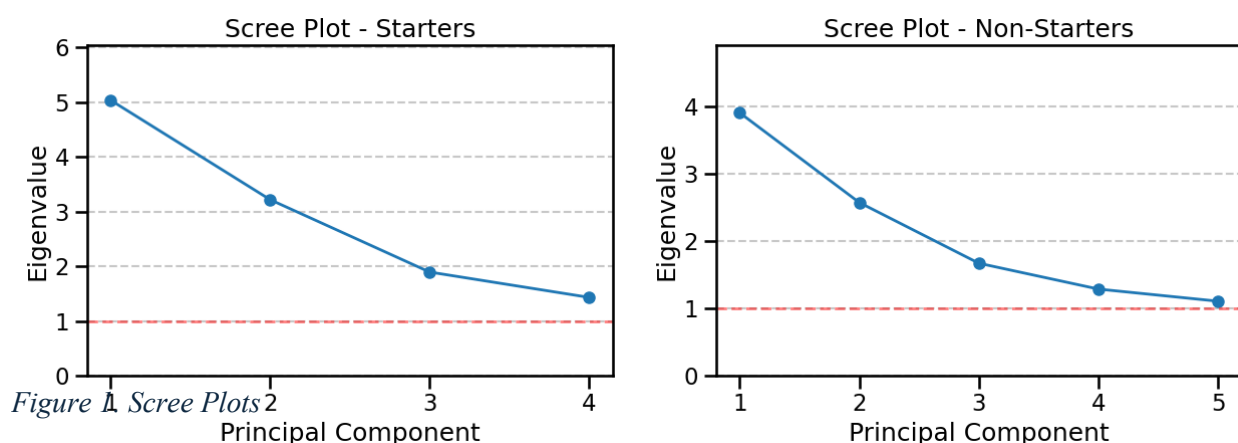
Statistical Design

Prior to statistical analyses, the data were separated into two groups: "Starters", consisting of the first athlete in each team, and "Non-Starters", consisting of athletes in positions 2, 3, and 4. Athletes from the 2025 event were excluded from the statistical analyses and reserved for out-of-sample testing. Statistical procedures were performed in Python version 3.12 using Pandas, Numpy, SkLearn, SciPy, Matplotlib, and Seaborn libraries. First, a correlation matrix was conducted and any variables with low correlation were removed from the dataset. Sampling adequacy was assessed using the Keiser-Meyer-Olkin test and Bartlett's test of sphericity. Next, a PCA was conducted, retaining principal components with eigenvalues greater than 1. Varimax rotation was applied to determine the loading for each of the principal components, similar to that of other research (Colyer et al., 2017; Jensen et al., 2023, 2023; Staunton et al., 2024b, 2024a). Variables with the strongest loading on each of

the principal components were retained for subsequent analysis. Then, stepwise multiple linear regression was conducted using the variables identified in the PCA. Standard estimate of error (SEE) was calculated for both regression equations. K-fold cross analysis was then used to assess the models' accuracy and stability (Colyer et al., 2017). SEE was calculated for the k-fold model and compared to that of the original. The correlation between k-fold predicted time and actual time was calculated and compared with the R^2 value of the original multiple regression. Lastly, 95% limits of agreement and 95% confidence intervals (CI) were calculated for the two models. Model accuracy and suitability were evaluated by calculating predicted times for each group using the corresponding regression equation applied to out-of-sample data. The agreement between the predicted and observed values was then visualized using Bland-Altman plots.

Results

In total, 80 athletes were analysed across 20 races. Following the previously discussed methods, a correlation matrix was conducted, and no variables were removed from either group. The KMO test resulted in 0.353 for the “Starters” and 0.356 for the “Non-Starters”. Bartlett’s test of Sphericity results showed p-values of <0.001 for both groups. PCA with varimax rotation yielded 4 principal components for the “Starters” and 5 for the “Non-Starters” groups, shown on *Table 2 Explained Variance & Component Loadings of the PCA*.



Principal Component	1		2		3		4		5	
Group	Starters	Non-Starters	Starters	Non-Starters	Starters	Non-Starters	Starters	Non-Starters	Starters	Non-Starters
Cumulative Variance (%)	26.28	29.37	59.47	48.71	73.1	61.27	83.45	70.95		79.28
Start Breakout Time	-	-	0.533	-	-	0.548	-	-		-
Rotate	-	-	-	-	-	-	0.764	-		0.837
Turn Breakout Time	-	-	0.474	-	-	0.632	-	-		-
Lap 1 Stroke Count	-0.404	0.506	-	-	-	-	-	-		-
Lap 2 Stroke Count	-	0.331	-0.430	-	-	-	-	-		-
FSS	-	-	-	0.563	-0.605	-	-	-		-
DPS	0.443	-	-	0.071	-	-	-	-		-
Stroke Rate	-0.444	0.504	-	-	-	-	-	-		-
Start15	0.344	-	-	-	-	-	-	0.678		-
Start15to25	-	-	-	-0.147	0.297	-	-	-		-
In5	-0.163	-	-	-	-	-	-	-0.164		-
Out15	-	-	-	-	0.235	-	-	0.653		-
Out15to25	-	-	-	-	0.451	-	-	-		-0.137
# of Variables	5	3	3	3	4	2	1	3		2

Table 2. Explained Variance and Component Loadings of the PCA

Bolded text represents the variables with the highest correlation in each component

“Starters” showed StrokeRate being most strongly associated with PC1 while “Non-Starters” was with Lap1StrokeCount. For PC2, “Starters” was most associated with StartTimeBreakout, and “Non-Starters” with FSS (Free Swimming Speed). PC3, showed “Starters” being the most strongly associated with FSS while the “Non-Starters” was with TurnTimeBreakout. PC4 showed the strongest association with Rotate in the “Starters” groups whereas Start15 was the strongest in the “Non-Starters”. The “Starters” did not have any more principal components with eigenvalues above 1 so analysis stopped. The last component in the “Non-Starters” PC5, showed the strongest association with Rotate. Based on the PCA results, variables with the strongest loadings were retained for regression analysis. Stepwise regression was then conducted to identify the key performance indicators for each group, resulting in the following regression equations (shown in tables 3 & 4). SEE was also calculated for each group and, following the k-fold validation, SEE and R²

differences were reported. Both groups did not exceed R^2 and adjusted R^2 differences of 0.10 when compared to the respective k-fold validation, deeming them stable (Kleinbaum et al., 2013; Rencher, 2002).

	Regression Equation	SEE	R^2	Adj R^2	r-value	p-value
Starters (Original)	Total Time = 80.142 + -16.455*FSS + 0.143*Rotate	0.27	0.71	0.67	0.84	3.43E-09
Starters (k-Fold)		0.25	0.78	0.72	0.88	2.02E-08
Difference		-0.02	0.07	0.06	0.04	1.68E-08

Table 3. Regression Results (Starters)

	Regression Equation	SEE	R^2	Adj R^2	r-value	p-value
Non-Starters (Original)	Total Time = 79.459 + -19.522*FSS + 1.258*Start15	0.155	0.940	0.937	0.970	9.60E-35
Non-Starters (k-Fold)		0.155	0.940	0.938	0.970	9.60E-35
Difference		0.00	0.00	0.001	0.000	0.00

Table 4. Regression Results (Non-Starters)

Bland-Altman plots were used to visualize the differences between the predicted and actual performances, both showing mean bias of <0.001, indicating a minimal prediction bias. The limits of agreement (LOA) for the “Starters” group were ± 0.498 s, with a 95% CI of ± 0.135 s, suggesting reasonable precision and limited random error. The “Non-Starters” group had a LOA of ± 0.297 s and a 95% CI of ± 0.047 s, indicating similar precision and random error to that of the other group. Overall, both groups showed strong agreement with actual swimming performance.

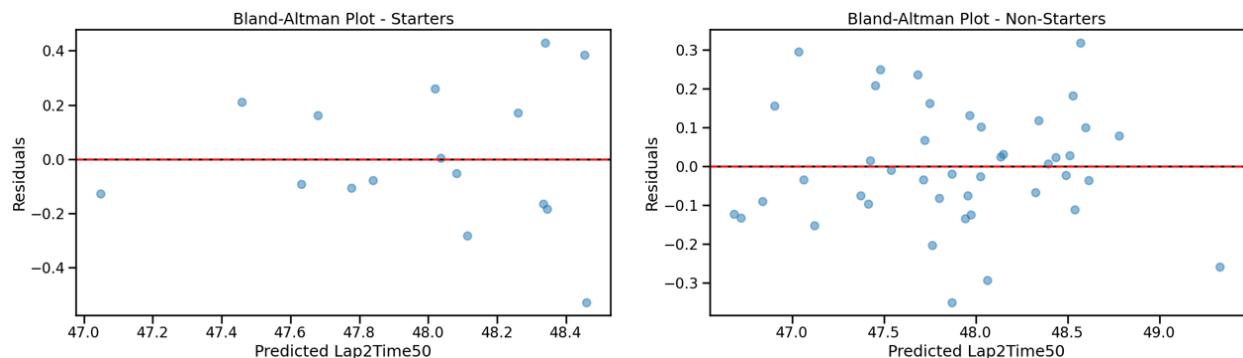


Figure 2. In-Sample Predictions

Out-of-Sample analysis showed imperfect prediction bias with a residual mean of 0.11s for the “Starters” and -0.07s for the “Non-Starters”. Root mean squared error (RMSE) was 0.149s and 0.061s for the respective groups, reflecting the average prediction error.

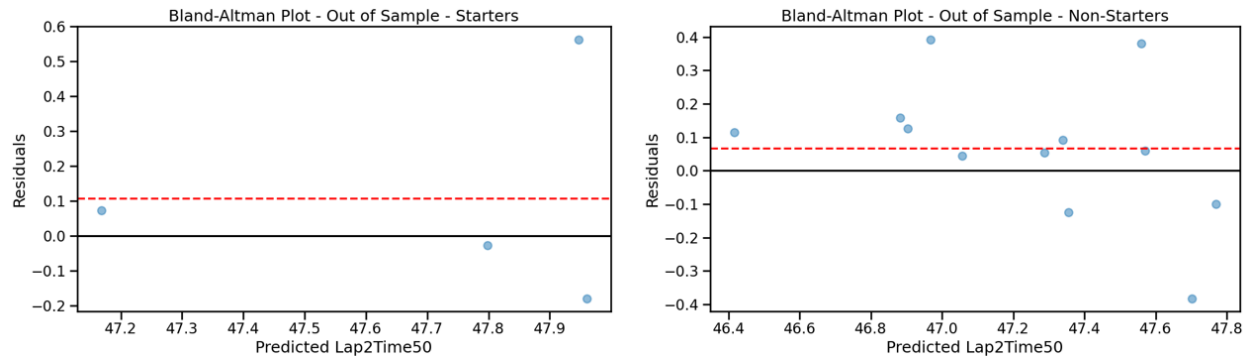


Figure 3. Out-Of-Sample Predictions

Discussion

The primary aim of this study was to identify key performance indicators (KPIs) for both roles of the men’s 4 x 100m freestyle relay and develop prediction models using PCA and multiple regression. These models were then evaluated for their stability and predictive accuracy to assess their potential utility for coaches and athletes. Bartlett’s test of sphericity was significant in both groups ($p < 0.001$), indicating that some relationships existed among the variables. However, low KMO values (< 0.5) indicate limited shared variance among variables, suggesting that the principal components are unlikely to represent coherent underlying constructs (Nkansah, 2018; Rencher, 2002). Consequently, the resulting regression formulas primarily reflect linear combinations of the original variables rather than meaningful dimensionality reduction.

It is important to mention that the identified KPIs in this study are based on their statistical significance in the applied methods. This does not imply that other metrics lack importance or contribution to performance, as race components play a role in successful

outcomes. Moreover, these metrics should not be viewed in isolation as targeted improvements in one phase may influence others and ultimately alter overall performance.

The cyclical and acyclic phases of swimming provide unique environmental variables and physiological demands. Stroke-related metrics such as FSS, DPS, and stroke count reflect a complex interplay between stroke rate and technique, likely producing optimal combinations for each athlete. For example, a stroke rate that is too high or too low may reduce DPS, resulting in a slower overall time. Relationships such as this one are inherently non-linear and may not be captured by linear regression models. Similarly, acyclic metrics such as Rotate may have diminishing returns, where improvements become progressively smaller beyond a certain point, potentially following a logarithmic pattern. For these reasons, future research using non-linear modelling may provide additional insights.

Although non-linear relationships may exist among these variables, linear regression models were used to quantify their predictive value. The resulting models for both groups were deemed stable as they had R^2 changes of less than 0.10 and strong statistical significance ($p < 0.001$). The “Starters” group showed moderate-to-strong proportion of variance, whereas the “Non-Starters” model explained a very high proportion of variance—a difference that likely influenced by the larger sample size of the “Non-Starters”.

The KPIs identified through regression were FSS and Rotate for the “Starters”, and FSS and Start15 for the “Non-Starters”. The inclusion of FSS as a KPI in both groups, and Start15 in the “Non-Starters” group, is broadly consistent with previous work done by Staunton et al., despite key methodological differences (gender, pool length, and events type) (2024a). It also aligns well Qui & Calvo’s work in 4 x 100m freestyle relays (2020). Interestingly, the identified KPIs from the “Non-Starters” group better matched the identified KPIs of Staunton’s work. Considering the methodological differences, and a different type of start, these KPIs may be more indicative of the stroke used in both events rather than

contextual factors. The differences between groups may also arise from sample size variations or strategic factors such as athlete order, rather than the type of dive technique used.

FSS has previously been identified as an important KPI across multiple strokes and distances (Born et al., 2021; Staunton et al., 2024b, 2024a). Because FSS broadly measures stroke performance—stroke rate (SR) or distance per stroke (DPS) may have been too specific to be individually identified as KPIs. It is important to note that while neither were identified, this does not diminish their importance—as they are large contributors to FSS.

The start and turn phases provide athletes with the unique opportunity to maximize force production by leveraging the wall or starting block. The identification of Start15 as a KPI in the “Non-Starters” group may reflect an athlete’s ability to generate speed off the block and sustain it to the 15m mark, serving as a proxy for performance across the acyclic phases of the race.

Rotate, identified in the “Starters” group, has previously been associated with performance in short-course competition, and its emergence in long-course competition aligns with prior findings, albeit in a different stroke (Born et al., 2021, 2022). Its relevance appears to increase with race duration, suggesting potential value in longer events such as the 4 x 200m freestyle relay (Born et al., 2021). In the sample groups, Rotate contributed $1.12\text{s} \pm 0.21\text{s}$ and $1.10\text{s} \pm 0.15\text{s}$ for the “Starters” and “Non-Starters”, respectively. Although this represents a relatively small proportion of total time compared to metrics such as Out15 ($6.75\text{s} \pm 0.11\text{s}$ and $6.84\text{s} \pm 0.15\text{s}$), its emergence as a KPI indicates that minor performance elements can have statistically significant impacts in elite competition. Given that Rotate reflects the speed and force of the turning action, it may indicate that athletes who turn faster are able to transfer power more effectively into the push-off, thereby optimizing use of the stretch-shortening cycle or displaying superior technique (Koster et al., 2025). This effect

may also be mediated by higher approach speed into the wall, acting as a proxy for a faster FSS.

Out-of-sample prediction showed the “Starters” group of having a mean bias of 0.11s with a range of -0.18s to +0.56s. “Non-Starters” showed a mean of -0.07s with a range of -0.38s to +0.39s. Given that differences of a few tenths of a second are meaningful in 100m freestyle, these error margins limit the models’ suitability as individual race-time predictors. Accordingly, the use of these models outside of training contexts may not be appropriate.

World-class athletes tend to compete in the same competition year-after-year. As such, many of the athletes in the out-of-sample group were also in the core sample group. This overlap likely inflates predictive accuracy relative to a truly independent sample. While this does reduce generalizability, it reflects a realistic competitive context, since many of the top athletes compete across multiple years. Caution is warranted when predictions are aggregated (e.g. estimating the time of a 4-man team), as any model error may be compounded, reducing accuracy at the team level. Future validation should use fully independent cohorts to better estimate the models’ accuracy on out-of-sample data. Although, producing a truly independent sample within this inclusion criteria may take years to achieve as athletes would need to retire from their current performance level or be replaced by faster athletes.

Practical Takeaways

The identified KPIs in both groups consist of FSS and a time-derived metric involving (Start15 or Rotate). Start15 can be monitored longitudinally without special equipment, as its calculation relies on standard race timing up to the 15m mark. While FSS and Rotate require additional split times between the 15-45m and the time at which the last stroke occurs.

Overall, data collection procedures remain straightforward and accessible for routine use—unlikely to pose as a barrier to success.

The smallest worthwhile change (SWC) in total time was 0.0913s for “Starters” and 0.1260s for “Non-Starters” (Bernards et al., 2017). Notably, the mean residual difference between out-of-sample observed and predicted total times was similar in magnitude to these SWC values for both groups, indicating that model error is comparable to the smallest worthwhile change. From the regression models, it is possible to estimate the change required in each KPI to illicit these thresholds. However, these estimates should be interpreted cautiously, as regression outputs do not always translate in a straightforward or intuitive way—particularly in the case of Rotate, as shown in *Table 5*.

Group	SWC = 0.2 x STD (Total time)	FSS	Start15	Rotate
Starters	0.0913s = 0.2 x 0.4563	*0.0056	-	*0.6451s
Non-Starters	0.1260s = 0.2 x 0.6301	*0.0065	*1.002s	-

Table 5. Smallest Worthwhile Change

* Change needed to reduce total time by the SWC (Smallest Worthwhile Change)

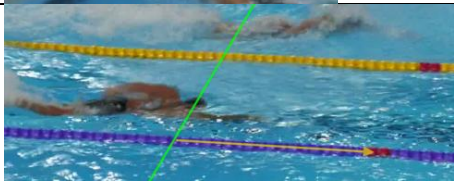
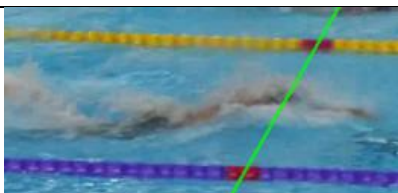
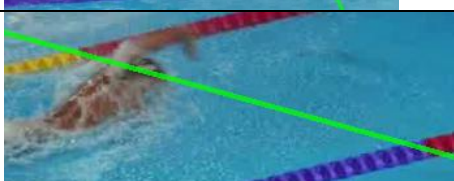
When applying the prediction models, practitioners should account for model bias and error. The outputs should be interpreted as indicative rather than definitive, particularly when used with non-elite athletes, where prediction accuracy may be reduced. Introducing additional variables such as exercise intensity is likely to reduce overall accuracy as it may introduce non-linear differences between variables—changing the proportions and variances of the variables within the data. As an example, a reduced intensity may not have as much of an impact on certain variables such as Rotate, so a cautious and holistic approach would be suggested when any modifications are made to the environmental variables.

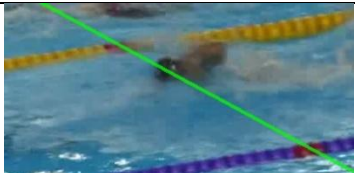
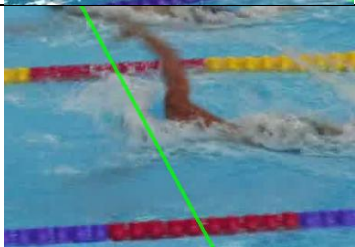
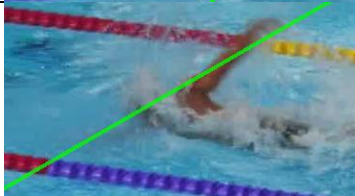
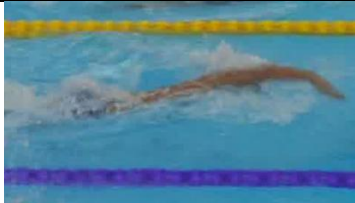
Conclusion

In conclusion, the main findings of this study highlight the important role of FSS and certain acyclic phase metrics (Start15 and Rotate) in the men's 4 x 100m freestyle relay. While PCA was applied, low KMO values indicate limited shared variance among the variables, suggesting that the principal components are unlikely to represent coherent underlying constructs. As a result, the regression models primarily reflect linear combinations of the original variables rather than true dimensionality reduction. Nonetheless, the stepwise multiple regression resulted in reasonable in- and out-of-sample performance. Limitations including sample overlap and the SWC being similar magnitude to measurement error suggest these models are best applied as training-support tools rather than definitive predictors of competition outcomes. Future work should validate these findings on fully independent samples, explore non-linear relationships, and utilize sub-elite athletes within the samples.

Appendix

Section 1. Video Analysis Definitions

Metric	Definition	
Finish Time	The Official time of the athletes 100m time	
RT	The time from the starting light (or teammates finish time) to when the athlete leaves the block	
Start Breakout Time	The time at which the athlete's arm leaves the water for their first stroke after the dive	
Start Breakout Distance*	The distance at which the breakout occurs	
Split 15m (0-50m)	The time at which the athlete's head passes the 15m mark of the 1st length	
Split 25m (0-50m)	The time at which the athlete's head passes the 25m mark of the 1st length	
Split 45m (0-50m)	The time at which the athlete's head passes the 45m mark of the 1st length	
Turn Start	The time of the hand entry for the athlete's last stroke in the length	
Split 50m (0-50m)	The Official time of the athletes 0-50m time	
Turn Breakout Time	The time at which the athlete's arm leaves the water for their first stroke after the turn	
Turn Breakout Distance*	The distance at which the breakout occurs after the turn	

Split 15m (50-100m)	The time at which the athlete's head passes the 15m mark of the 2nd length		
Split 25m (50-100m)	The time at which the athlete's head passes the 25m mark of the 2nd length		
Split 45m (50-100m)	The time at which the athlete's head passes the 45m mark of the 2nd length		
Split 50m (50-100m)	The Official time of the athletes 50-100m time		
Stroke Count	Each hand entry for each stroke is recorded based on its relative time to the race		

** Distances were derived from their distance to the nearest landmark (represented by the yellow arrow)*

Section 2. Reliability Testing

	Finish Time	RT
AVG	0.091%	0.000%
STD	0.429%	0.000%

	Start Breakout Time	Start Breakout Distance	Split 15m (0-50m)	Split 25m (0-50m)
AVG	0.024%	1.824%	0.876%	0.436%
STD	0.080%	3.575%	3.993%	2.028%

	Split 45m (0-50m)	Split 50m (0-50m)	Turn Start	Turn End
AVG	0.215%	0.000%	0.397%	0.000%
STD	0.992%	0.000%	1.307%	0.000%

	Turn Breakout Time	Turn Breakout Distance	Split 15m (50-100m)	Split 25m (50-100m)
AVG	0.012%	4.948%	0.007%	0.230%
STD	0.041%	5.017%	0.014%	0.737%

	Split 45m (50-100m)	Split 50m (50-100m)
AVG	0.094%	0.091%
STD	0.406%	0.429%

*Table 6. Reliability Testing
Represented as % Error (Hughes, 2004)*

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